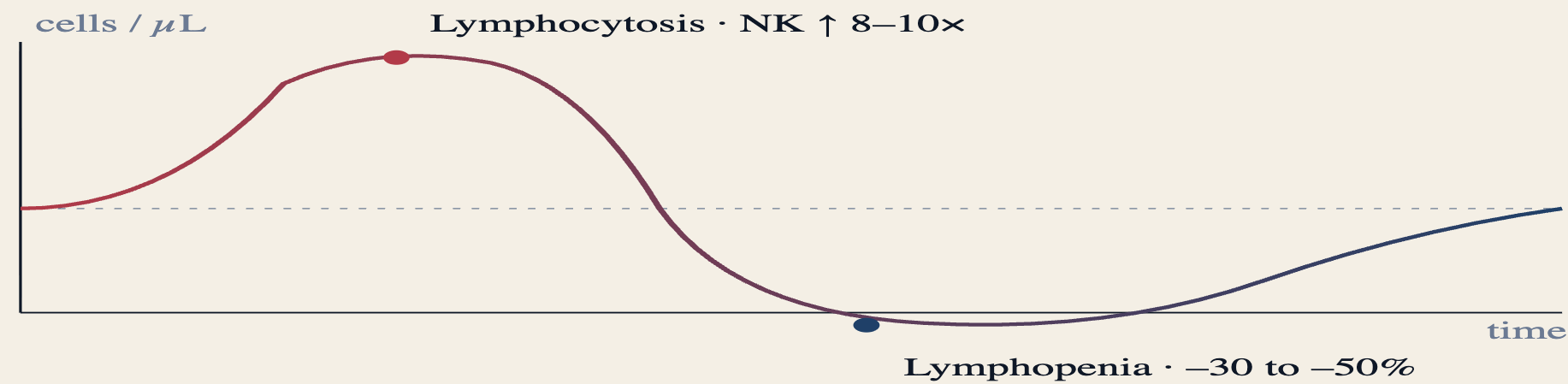


LECTURES 2 & 4 — TWIN SUMMARY DECKS

The biphasic immune response & IL-6 as the prototypical *myokine*.



SOURCE

[chrisputa.github.io / exercise_immunology](https://chrisputa.github.io/exercise_immunology) Lectures 2 & 4

FORMAT

2 $\times$ 10 slides — twin decks

TOPICS

Lymphocyte trafficking · catecholamines · IL-6 · myokine
signaling · Ringleb meta-analysis

AUTHOR

After Puta, Haunhorst & Gabriel (2026)

USE $\leftarrow$ $\rightarrow$ TO NAVIGATE

The biphasic immune cell response to *acute dynamic exercise*.

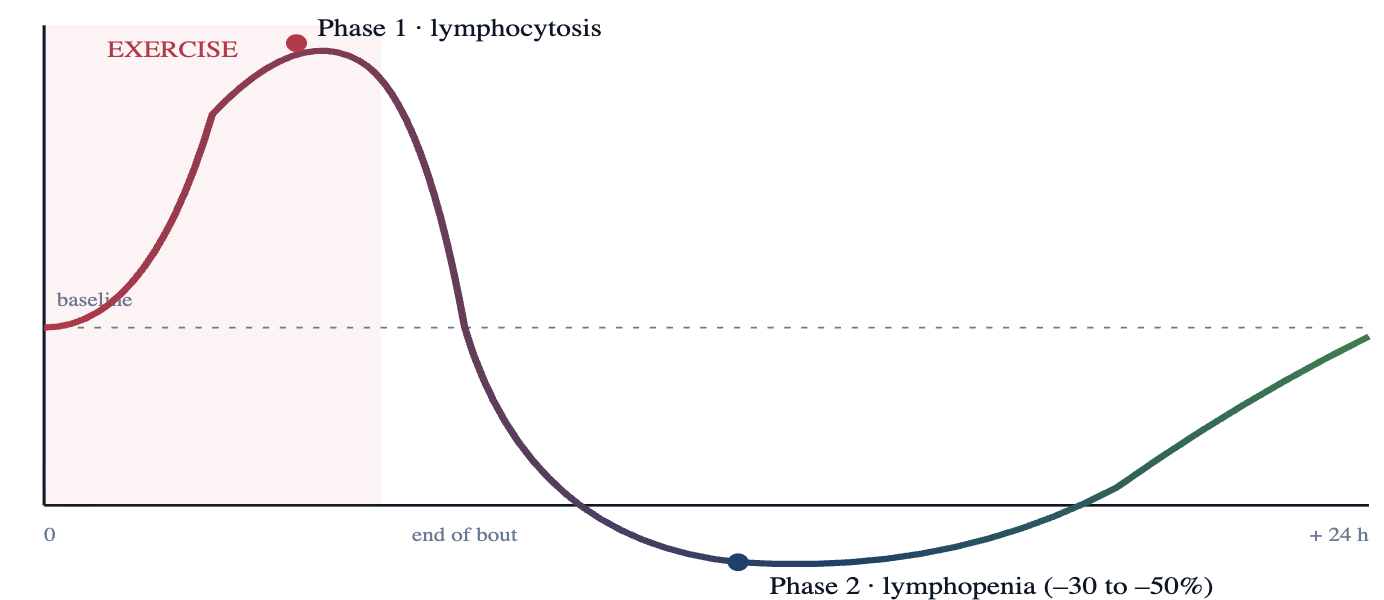
A 10-slide tour of how a single bout of running, cycling or interval work moves lymphocytes in and out of circulation — and why that movement is surveillance, not suppression.

A single hard bout produces *two* distinct immune phases — not one.

Acute dynamic exercise of $\geq 20\text{--}60$ min , or high-intensity intermittent work, drives a stereotyped immunological response — most clearly visible in the lymphocyte compartment.

During the bout, blood is flooded with cytotoxic cells. After it ends, those same cells disappear from circulation for up to 24 h. Both arms together form a *surveillance loop*, not the immunosuppression older textbooks described.

SCHEMATIC — LYMPHOCYTE COUNT VS. TIME



Cytotoxic cells flood the blood within minutes.

NK CELLS

8–10x

above baseline. The dominant signal of Phase 1.

CD8⁺ & $\Gamma\Delta$ T CELLS

2–3x

Effector-memory subsets are mobilized preferentially over naïve cells.

NEUTROPHILS

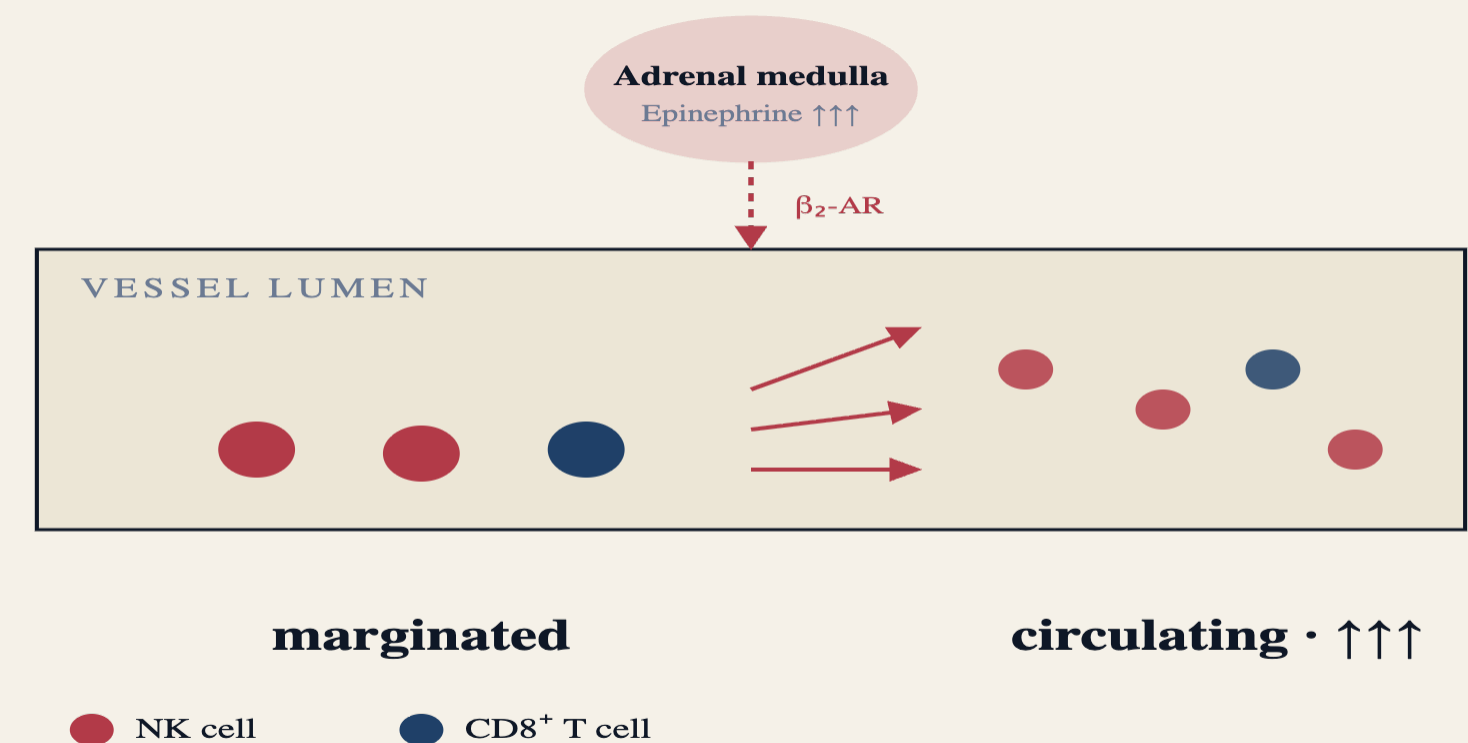
~2x

Demargination from vessel walls; oxidative burst potentiated at moderate loads.

*Acute exercise preferentially mobilizes **CD8⁺ T-cell subsets with anti-tumor transcriptional profiles** and **CD3⁺CD56⁺ NK-T-like cells** — the cells you'd most want patrolling tissues.*

Epinephrine pulls cytotoxic cells off the vessel wall via β_2 .

- 01 **Epinephrine surges** with exercise intensity — the chief driver of mobilization.
- 02 Catecholamines bind **β_2 -adrenergic receptors** on lymphocytes, decreasing their vascular adhesion.
- 03 Increased **cardiac output and shear stress** physically dislodges marginated cells.
- 04 Mobilized NK cells use the same effector machinery — **granzyme/perforin and TRAIL** — and exhibit increased per-cell cytotoxicity.



After the bout: cells *redistribute* to peripheral tissues.

NADIR VS. BASELINE

–30 to –50%

circulating lymphocytes, reached **45 min – 2 h** after exercise stops; resolves within ~24 h.

Mobilized cells don't die — they traffic to peripheral target tissues (lung, gut, skin, BM).

At the same time, **cortisol-driven neutrophilia** rises in the blood. Modern

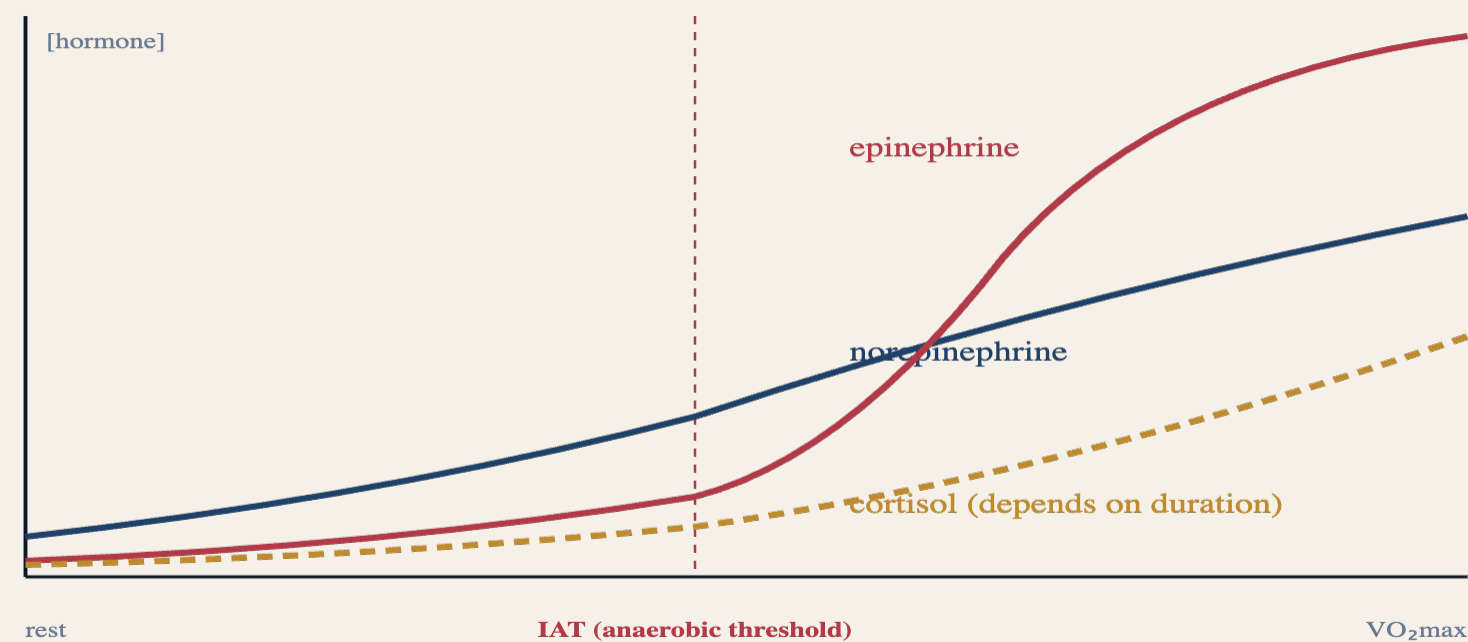
interpretation: a window of **heightened immune surveillance** , not a "open window" of vulnerability.

WHERE THE CELLS GO

- **Lung & gut mucosa** — front-line surveillance.
- **Bone marrow** — recycling and re-licensing of effector cells.
- **Tumor / infected tissue** — anti-tumor CD8 transcriptomes are enriched.
- **Skin & injured muscle** — $\gamma\delta$ T cells and NK-T-like cells home to barrier sites.

*"Lymphopenia = **surveillance** , not suppression." — Campbell & Turner, Front Immunol 2018.*

The IAT is the *breakpoint* — physiological, hormonal, immunological.



EPI

Above the **individual anaerobic threshold (IAT)**, epinephrine rises **over-proportionally**. This is the hormonal & immunological inflection point.

NE

Norepinephrine scales more linearly with intensity — a sympathetic-tone signal.

COR

Cortisol depends on **both intensity and duration**. In ultra-endurance, increases can exceed **10x** baseline.

*Implication: prescribing exercise relative to IAT lets you **predict the immune response**, not just the metabolic one.*

Below threshold: nothing happens. Above it: everything.

SUBTHRESHOLD ZONE

$\leq 100\%$ IAT for ≤ 30 min or extensive endurance ≤ 1 h

Virtually **no influence** on immune cell trafficking in the post-exercise phase.

Moderate work even **stimulates** the neutrophil oxidative burst.

SUPRATHRESHOLD ZONE

Intense intervals · exhaustive continuous work

HIIT sessions produce immune responses **comparable to exhaustive continuous runs**. Intense exercise **suppresses** the neutrophil oxidative burst.

A practical rule of thumb:

A Easy / threshold work → metabolic adaptation, minimal immune trafficking.

B Sustained suprathreshold → full biphasic response.

C HIIT → biphasic response in a fraction of the time, via the same epinephrine surge.

D Ultra-endurance → intensity is modest but cortisol $> 10\times$ baseline reshapes the curve.

Repeating the bout reshapes the resting baseline.

INFLAMMATION



Lower basal pro-inflammatory cytokines . Skeletal muscle becomes a chronic source of anti-inflammatory IL-6 / IL-1RA / IL-10 signaling.

NK CYTOTOXICITY



Higher per-cell killing capacity at rest. Improved tumor and viral surveillance.

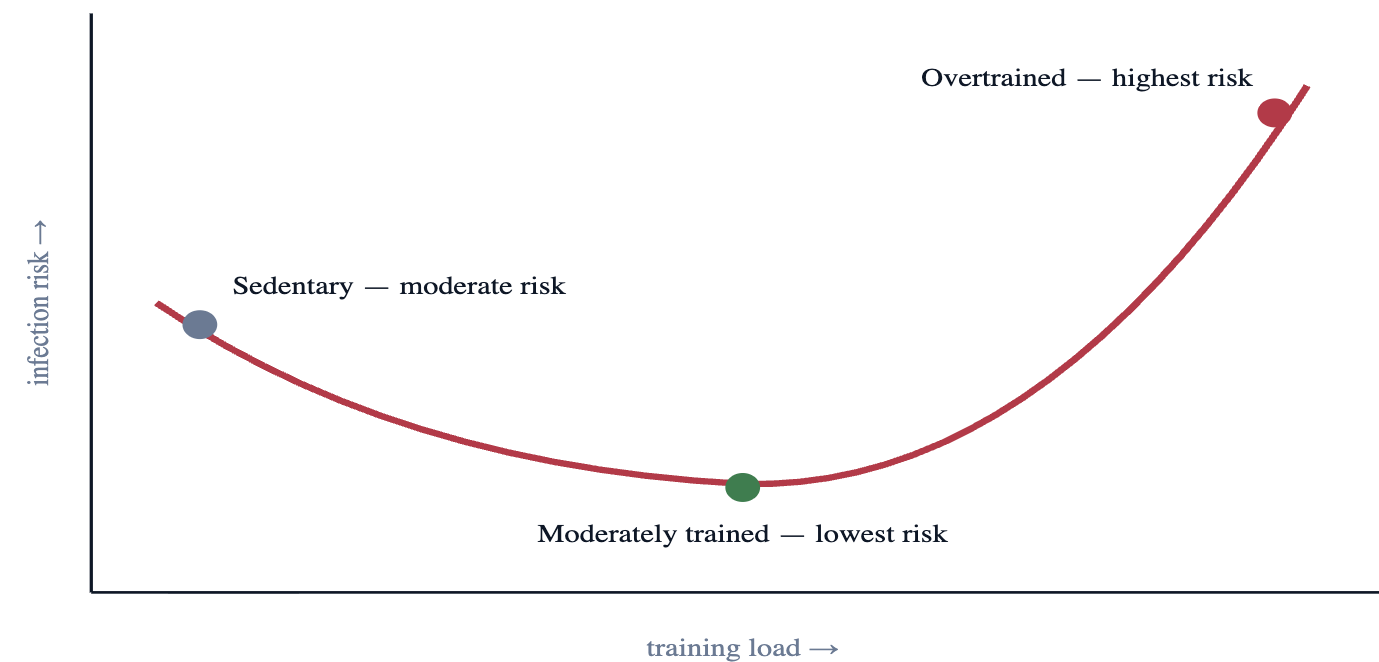
T-CELL SENESENCE



Reduced T-cell senescence and greater clonal diversity — a younger immune repertoire across the lifespan.

*Muscle-derived IL-6 can rise **up to 100x above basal** during contraction. Unlike immune-derived IL-6, it triggers an **anti-inflammatory** cascade (↑ IL-1RA, ↑ IL-10, ↓ TNF). The same molecule, opposite role — this is the bridge to Lecture 4.*

Infection risk vs. training load: a *J*, not a line.



- PRO** Survey data on URTI incidence in elite endurance athletes around competition supports the upturn at very high loads.
- CON** Self-reported "infection" often reflects **airway inflammation, not pathogenic infection** — symptoms ≠ illness.
- CON** Many studies are **cross-sectional, single-bout, biased** toward male elite cohorts.
- USE** Useful as a teaching frame; should be applied **critically, not literally**.

"Debunking the myth of exercise-induced immune suppression" (Campbell & Turner, 2018) is the modern counter-text.

Five things to take with you.

01 · TWO PHASES

Acute exercise → **lymphocytosis** (Phase 1) → **lymphopenia** (Phase 2). Both are normal, both are surveillance.

03 · β_2 DOES THE MOVING

Epinephrine on β_2 -adrenergic receptors demarginates cells. Cortisol shapes the lymphopenic arm.

05 · TRAINED ≠ UNTRAINED AT REST

Regular activity lowers basal inflammation, raises NK cytotoxicity, slows T-cell senescence.

02 · NK CELLS LEAD

NK cells rise **8–10x** ; CD8⁺ effector-memory and $\gamma\delta$ T cells follow at 2–3x.

04 · IAT IS THE SWITCH

Below the individual anaerobic threshold, immune trafficking is silent. Above it, the full curve appears.

06 · RETHINK THE J-CURVE

Useful sketch, contested data. Phase 2 is **surveillance** , not an "open window".

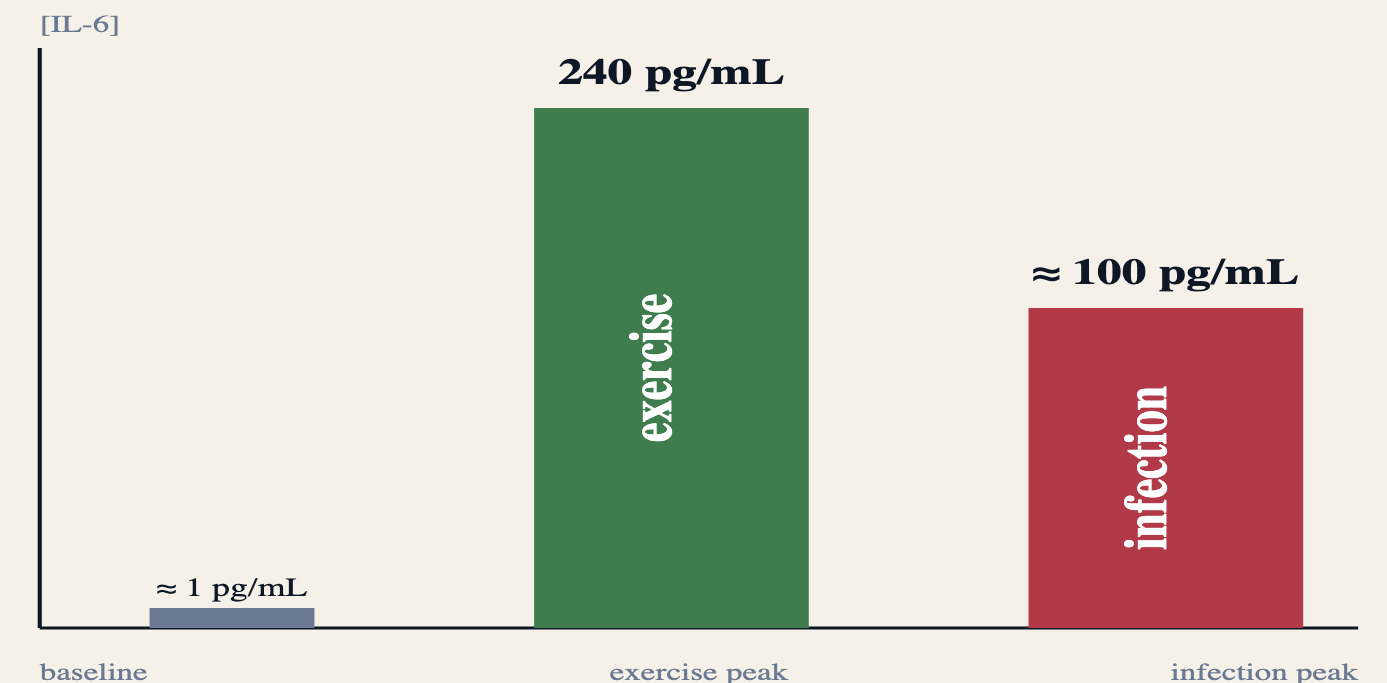
IL-6 as the prototypical *myokine* : cytokine context determines function.

Same molecule, opposite meaning. Why exercise-induced IL-6 builds health, while infection-induced IL-6 damages tissue — and how the Ringleb 2026 meta-analysis quantifies it.

IL-6 hits *240 pg/mL* in exercise — and the body gets *healthier*.

For decades IL-6 was filed as a pro-inflammatory cytokine because of its role in infection. Yet during a 60-min run it reaches concentrations that should — by that logic — be devastating.

The resolution is that IL-6 is **fundamentally context-dependent**. Its biological function is set not by its concentration but by the surrounding cytokine milieu — specifically, the presence or absence of **TNF- α** and **IL-1**.



*"IL-6 functions as a **metabolic signaling molecule** in exercise — not an inflammatory amplifier — because the upstream pro-inflammatory signals are absent." — Kistner, Pedersen & Lieberman 2022.*

The fingerprint is in *what comes with* IL-6.

INFECTION CASCADE Pro-inflammatory amplifier TNF-α rises first (minutes–hours), activates NF-κB. IL-1 follows, perpetuating signaling. IL-6 appears after 4–8 h as a secondary amplifier. Counter-regulation (IL-1ra, IL-10) arrives late (6–8 h, 4–6 h). Sustained: TNF > 10 h, IL-6 24–48 h or longer.	EXERCISE CASCADE Metabolic signal TNF-α and IL-1 are absent — the diagnostic feature. IL-6 rises directly from contracting muscle. IL-1ra and IL-10 rise concurrently (within 2–3 h). Active anti-inflammatory milieu coexists with very high IL-6. Resolves in 24–48 h — "biphasic with immediate resolution".
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*The most discriminatory marker: exercise produces **IL-1ra without producing IL-1** . This proactive antagonism is immunologically unique to muscle contraction.*

MAGNITUDE VARIES — PREDICTABLY

Endurance running tops the table. Resistance work, the floor.

EXERCISE TYPE	DURATION	INTENSITY	PEAK IL-6 (PG/ML)	TIME TO PEAK	RETURN TO BASELINE
Endurance running	60 min	75% $\dot{V}O_2$ max	240	2–3 h	24–48 h
HIIT	30 min	maximal intervals	165	1.5–2.5 h	24–48 h
Moderate continuous	45 min	60–70% $\dot{V}O_2$ max	85	1.5–2 h	24–48 h
Resistance exercise	30 min	moderate volume	42	2–3 h	24–48 h

Whole-body endurance produces **substantially larger** myokine responses than lower-body-only modalities at equal intensity — a **muscle-mass** argument.

Mechanistically: more recruited muscle → more glycogen depletion → higher AMP/ATP ratios → stronger AMPK signal → larger IL-6 transcriptional pulse.

Sense the deficit. Liberate the fuel. Allocate it to muscle.

A · ENERGY SENSING

Read the muscle.

Lactate accumulation, ROS, glycogen depletion → AMPK activation → PGC-1α / SIRT1 pathways → **IL-6 transcription** . Not an NF-κB program.

B · ENERGY LIBERATION

Open the stores.

In circulation, IL-6 drives **hepatic gluconeogenesis** and **adipose lipolysis** ; indirectly induces cortisol secretion to amplify catabolism.

C · ENERGY ALLOCATION

Route to muscle.

Increases **insulin sensitivity**, **GLUT4 expression** , and **fatty-acid transporter expression** in muscle.
Transiently down-regulates competing energy uses — including immune activity.

*The "anti-inflammatory" effect of exercise is a **side-effect of energy allocation** : IL-6 transiently dampens immune activity while shunting fuel to working muscle.*

Two routes to the IL-6 gene. Only one of them is *inflammatory*.

INFECTION ROUTE

TNFR1 → NF-κB → IL-6

Sustained NF-κB activation in immune cells, endothelium, fibroblasts. **Elevated TNF receptor**, sustained IL-6, **delayed counter-regulation**.

EXERCISE ROUTE

↑AMP/ATP → AMPK → PGC-1α → SIRT1 → IL-6

Minimal NF-κB activation despite IL-6 concentrations **100–240× baseline**. TNF receptor only $\sim\frac{1}{3}$ – $\frac{1}{2}$ the level seen in infection.

The **IL-1ra without IL-1** pattern is the immunological fingerprint:

INF

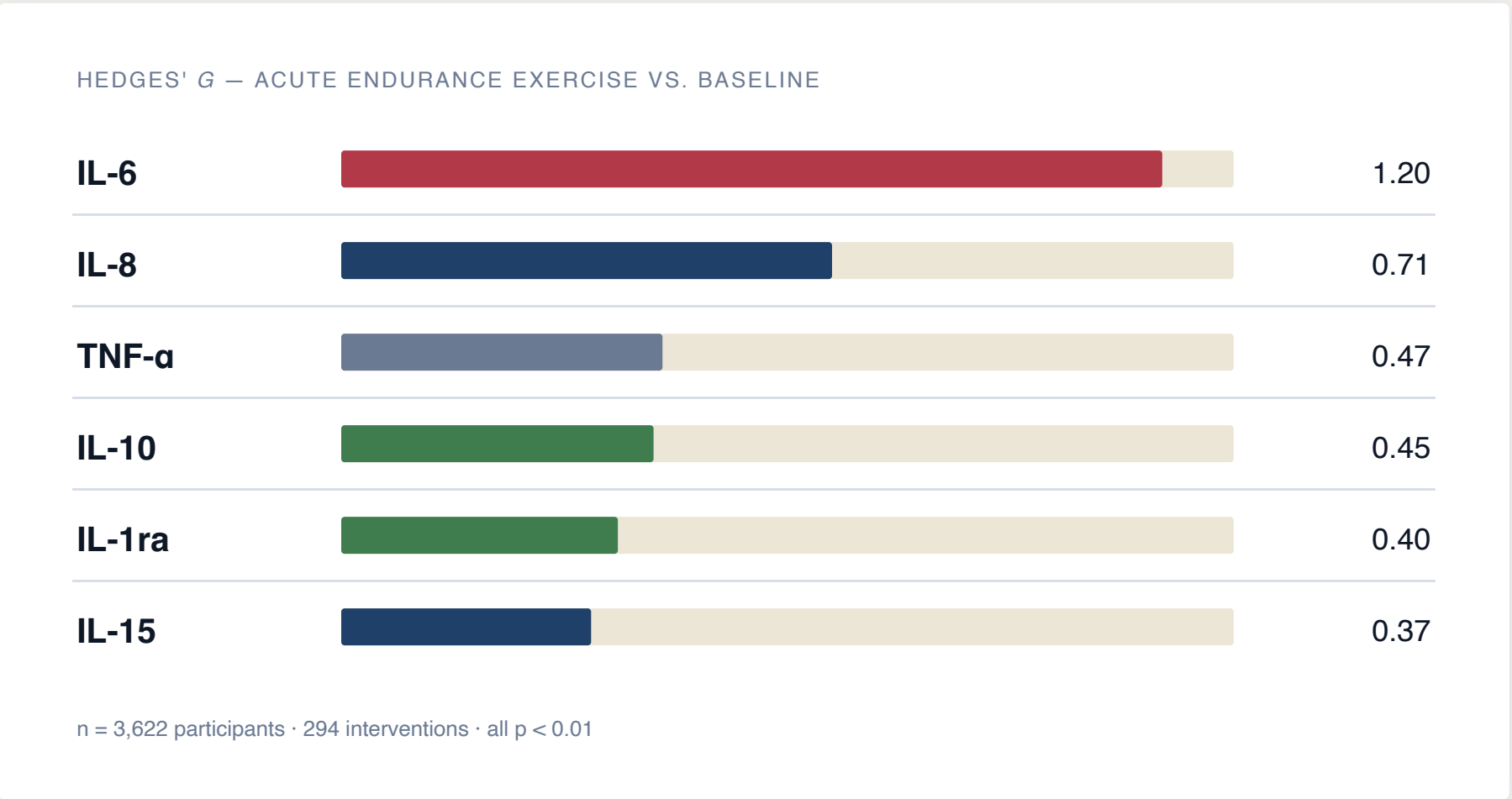
IL-1 produced first → tissue damage begins → IL-1ra arrives **late** as inadequate brake.

EX

IL-1ra produced **without antecedent IL-1**, concurrent with IL-6. Any residual IL-1 is neutralized before it amplifies.

*High IL-6 + high IL-1ra + **no** IL-1 + **no** TNF = the unique signature of a healthy myokine response.*

IL-6 dominates. The others orbit it.



XL IL-6 effect size **g = 1.20** — a *very large* effect, dwarfing the others.

M Counter-regulators IL-10 / IL-1ra at **g ≈ 0.4** — modest but consistent.

Ø TNF-α at **g = 0.47** — present but small, in line with the Kistner "no NF-κB" prediction.

*Quantitative confirmation that IL-6 is the **prototypical myokine** — and that it travels with its own counter-regulators, not with TNF/IL-1.*

Exercise dose explains ~44% of IL-6 variance.

DURATION ALONE

39%

of IL-6 variance (R^2 , $p < 0.001$).

Each additional **15 min of moderate-intensity endurance** predicts ~0.25–0.30 SD increase in post-exercise IL-6. This validates the **energy-sensing** mechanism: dose tracks glycogen depletion tracks AMPK activation tracks IL-6.

FULL DOSE MODEL

44%

when intensity × duration combined.

RUNNING VS. CYCLING — HEDGES' G

MYOKINE	RUNNING	CYCLING	P
IL-6	1.30	1.10	n.s.
IL-10	0.63	0.24	< 0.01
IL-1ra	0.52	0.21	< 0.01
TNF-α	0.65	0.36	< 0.01

~2.5× larger IL-10 response with running — a **muscle-mass** dividend for whole-body activity.

Match the dose to the goal. Watch IL-6's *shape*, not just its level.

GOAL	MODALITY	DURATION	INTENSITY	EXPECTED EFFECT
Max anti-inflammatory	Running	45–60 min	75–90% $\dot{V}O_2$ max	IL-6 $g \approx 1.30$ · IL-10 $g \approx 0.63$
Moderate	Cycling / mixed	30–45 min	60–75%	IL-6 $g \approx 1.10$ · IL-10 $g \approx 0.24$
Time-limited / orthopedic	HIIT / cycling	20–30 min	maximal intervals	IL-6 $g \approx 1.15$ · IL-8 $g \approx 0.71$

"Even suboptimal exercise conditions can elicit significant increases in immunoregulatory myokines" — meaning **any sustained endurance bout** produces a meaningful response, even when ideal dosing isn't possible.

NORMAL ADAPTATION

High IL-6 ($g > 0.8$)

Concurrent $\uparrow$ IL-1ra / IL-10

TNF- α normal to mildly $\uparrow$

Resolves in 24–48 h

OVERTRAINING RED FLAGS

IL-6 still elevated **at 48–72 h**

Delayed or absent IL-1ra / IL-10

Sustained TNF- α — NF- κ B engagement

Reduce load, intensify recovery

Five things to take with you.

01 · CONTEXT, NOT CONCENTRATION

IL-6 means different things in muscle vs. infection. The **cytokine milieu** sets its function.

03 · ENERGY ALLOCATOR

IL-6 senses, liberates, and allocates energy — the anti-inflammatory effect is a side-product of fuel routing.

05 · RINGLEB CONFIRMS IT

IL-6 g = 1.20, dose explains 44% of variance, running > cycling — the framework holds quantitatively.

02 · THE FINGERPRINT

Exercise IL-6 travels **without TNF/IL-1, with concurrent IL-1ra and IL-10** — the diagnostic signature.

04 · AMPK, NOT NF-KB

Glycogen depletion → AMPK → PGC-1α → SIRT1 → IL-6. Different gene-regulatory route → different outcome.

06 · WATCH THE SHAPE

Sustained IL-6 with delayed IL-1ra/IL-10 is the **overtraining** signature. Use the curve, not the value.